

Buck-Boost Converter Report

EE252: EMPEL (B1 - G3)

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Problem Statement

To analyze Buck and Boost converter operation by observing switch current, inductor current, and inductor voltage waveforms in CCM and DCM.

Specifications

Input: 15 V

Output: 10–20 V Switching Frequency: 7.5 kHz

Output Current: 1 A

Experimental Values

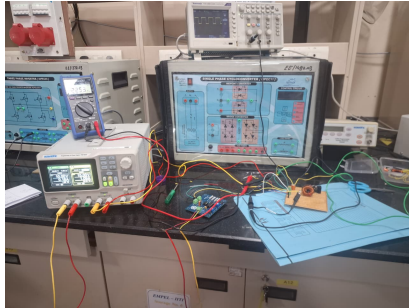
Buck Mode:

- Input Voltage (V_{in}): 15 V
- Output Voltage (V_o): -7.33 V
- Load Resistance (R): 8 Ω
- Duty Cycle (D): 0.37
- Input Current (I_{in}): 0.68 A
- Efficiency: 0.6584

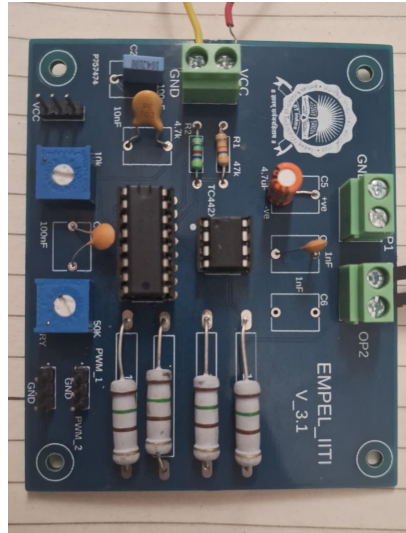
Boost Mode:

- Input Voltage (V_{in}): 15 V
- Output Voltage (V_o): -17.1 V
- Load Resistance (R): 23 Ω
- Duty Cycle (D): 0.62
- Input Current (I_{in}): 1.23 A
- Efficiency: 0.6891

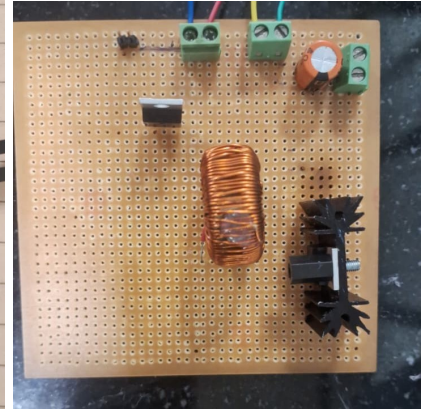
Hardware Setup



(a) Setup



(b) Driver



(c) Converter

Figure 1: Hardware Images

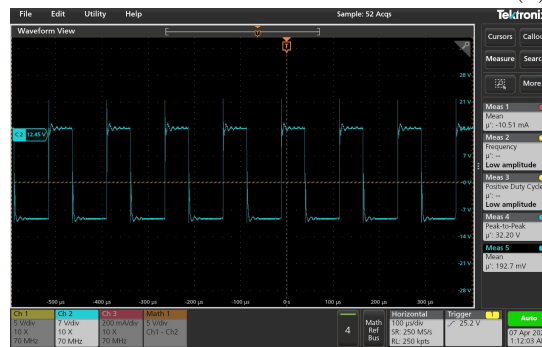
Buck Mode (CCM)



(a) I_L



(b) I_{SW}



(c) V_L

Figure 2: Buck CCM Waveforms

Boost Mode (CCM)



(a) I_L

(b) I_{sw}



(c) V_L

Figure 3: Boost CCM Waveforms

DCM Operation



Figure 4: DCM

Conclusion

The Buck-Boost converter operates in CCM at higher switching frequency where inductor current is continuous. When switching frequency is reduced, the converter enters DCM where inductor current becomes discontinuous, validating theoretical predictions.